8 AI and Politics

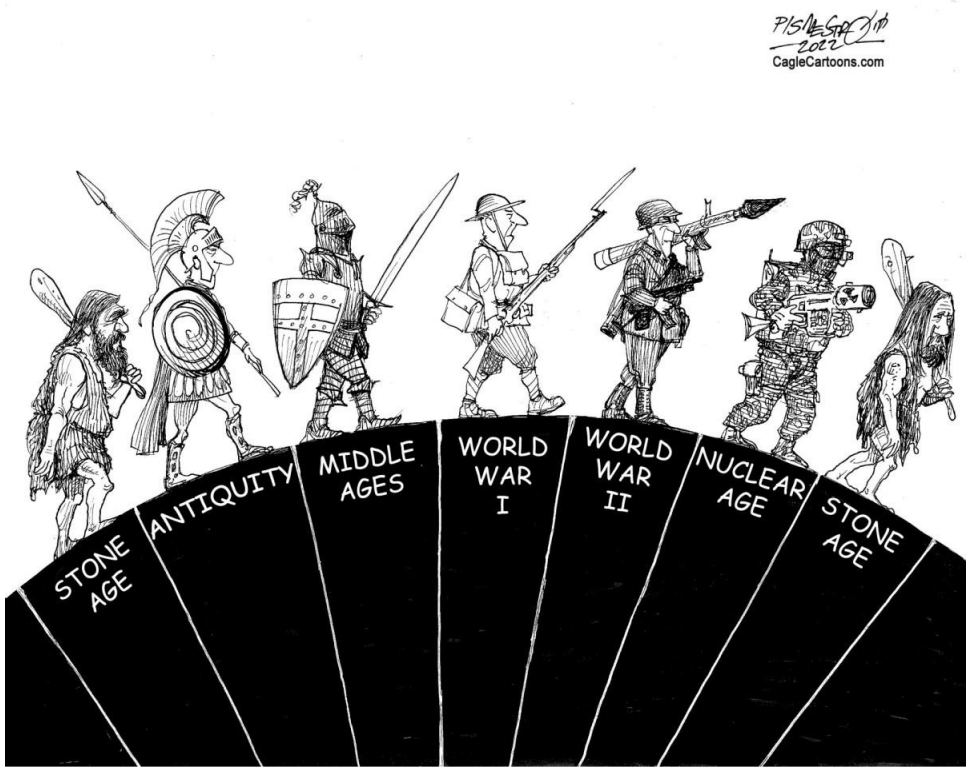
GEB2508 AI and Society

Review of Last Class

- Structure and Agency
  - Technological Neutrality Thesis
  - Agency and Responsibility
- Human-Machine Relation
  - Cyborg and Posthumanism
- Socio-Technical System
  - Technology Diffusion and Substitution
  - Path Dependence & Lock-in
  - AI as Socio-Technical Infrastructure

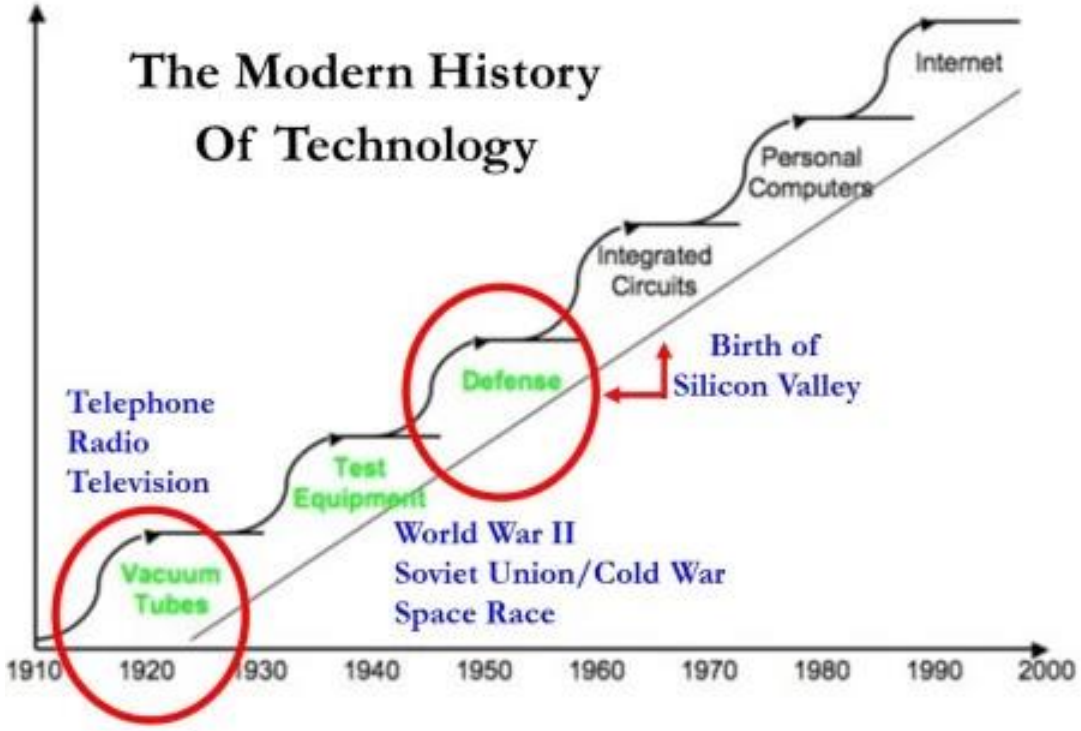
Content

- Warfare and Technology
  - Cold War: Arms Race and Space Race
  - Military-Industrial Complex
  - Autonomous Weapons and Cyberwarfare
- Nation-State and Geopolitics
  - State Theory
  - Industrial Policy and Techno-nationalism
  - AI Arms Races and Geopolitics



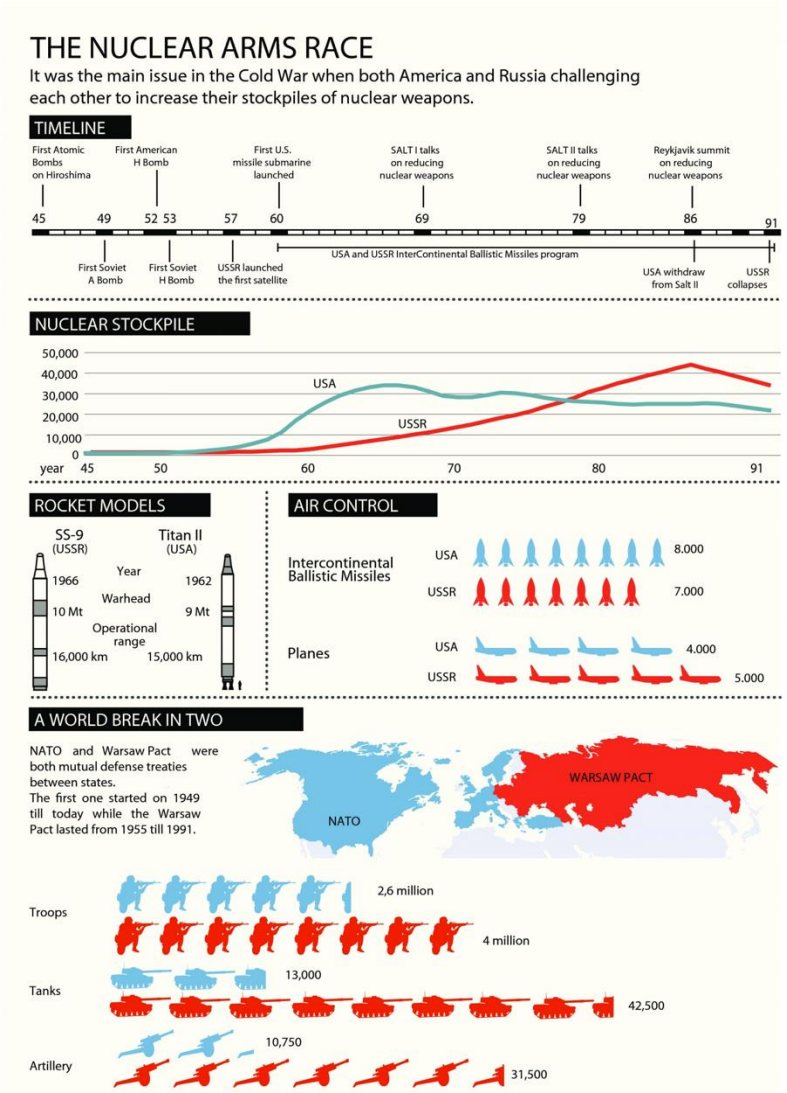
Relationship Between Technology and Warfare

- Technology as an influential forces shaping warfare
  - Gunpowder, the steam engine, information technology, etc.
- Warfare as a **driver** of technological innovation
  - Dual-use technologies and civilian spillovers: internet, digital cameras, ambulance etc.



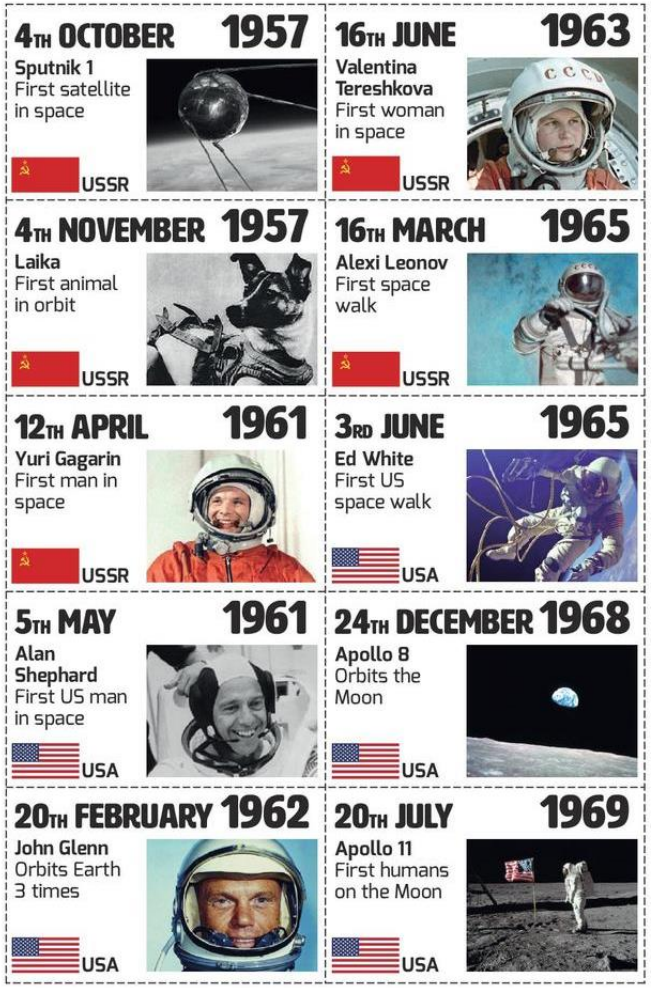
Cold War (1947-1991)

- Escalation: technological advancements often lead to arms races
- Nuclear strategy: **deterrence**
  - If one side attacked first, it knew that the other could retaliate.
- Mutually assured destruction (MAD)
  - "I know not with what weapons WWIII will be fought, but WWIV will be fought with sticks and stones." - Einstein



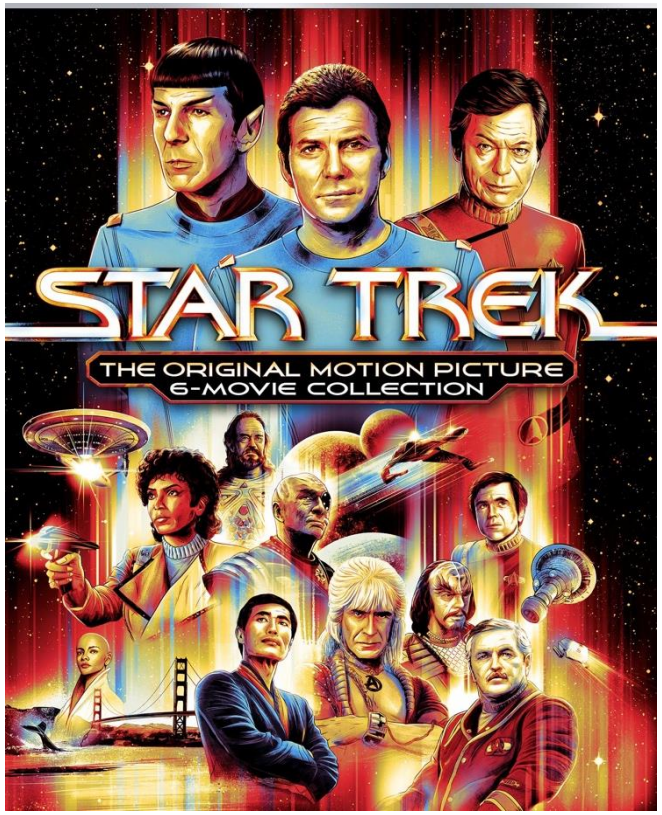
Space Race (1957-1975)

- Space achievements as proof of **ideological** superiority
  - Soviet framing vs. US framing
- Space technology in everyday life
  - E.g., GPS, digital photography, laptops, solar panels, LED, etc.
- Model of "**big science**": tight government-industry-academia collaboration

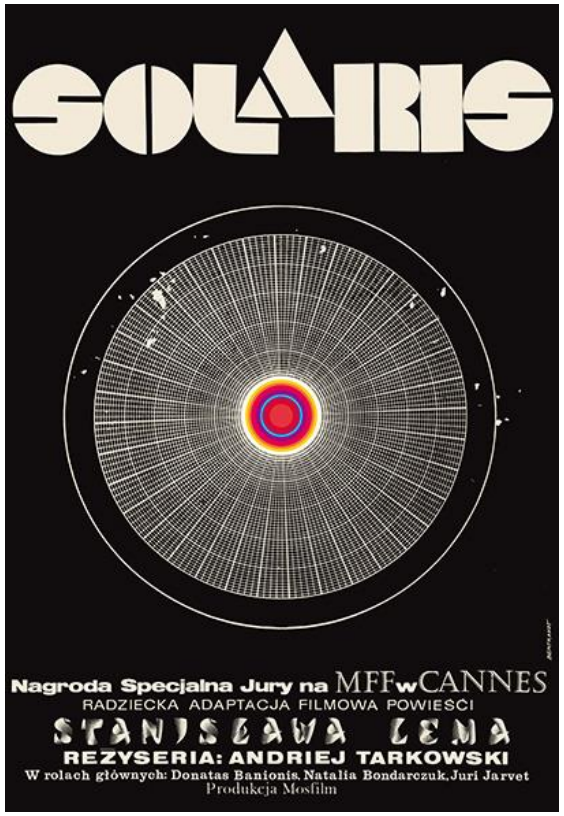


Space Race in Popular Culture

Star Wars (1977-)



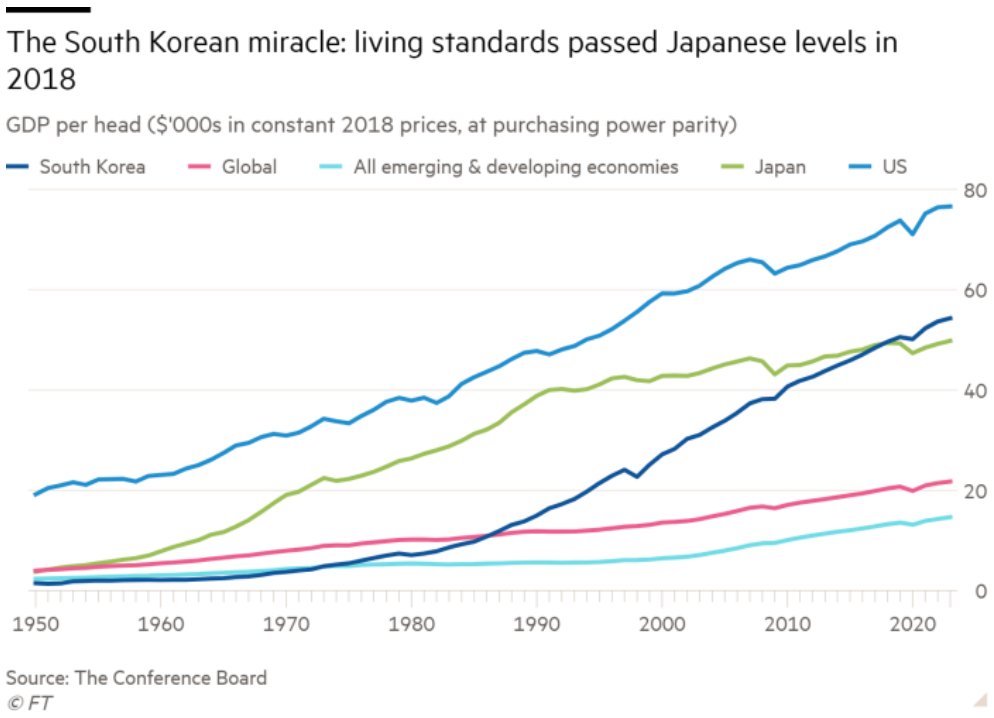
Star Trek (1966-)



Solaris (1972)

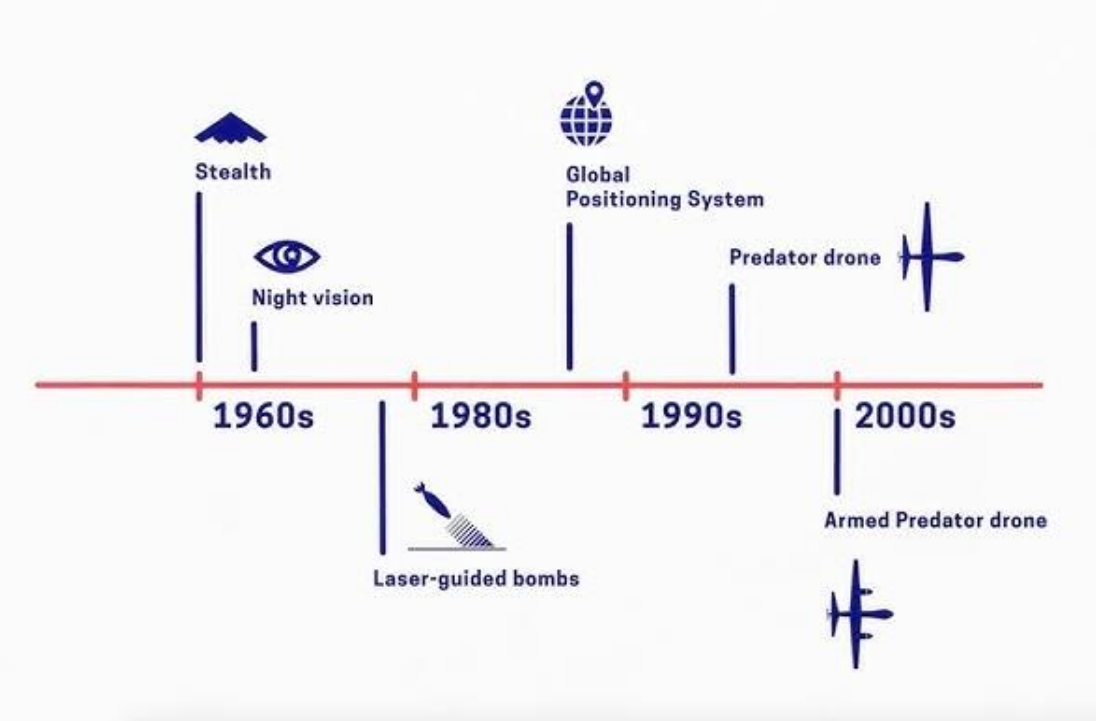
Geopolitical Ramifications

- The Cold War provided the geopolitical context for the East Asian **industrialization**.
  - U.S. economic and military aid
  - Export-oriented industrialization
- Japan and the Korean War (1950-1953)
- South Korea and the Vietnam War (1955-1975)



Technology and Transformation of Warfare

- Gunpowder Revolution (14–17C)
  - Cannons, firearms, fortification
- Industrial Age (18–19C)
  - Railroads, telegraphs, naval warfare
- Mechanized World Wars (20C)
  - World War I: airplanes, tanks, and submarines;
  - World War II: radar, aircraft carriers, nuclear weapons

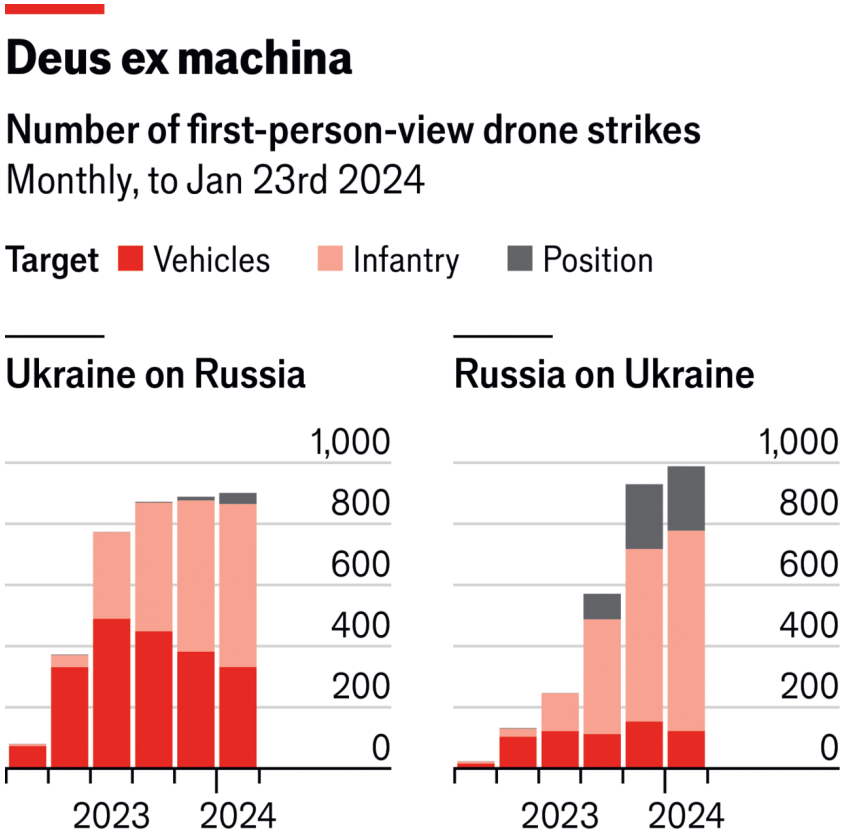


AI and Autonomous Weapon Systems

- Unmanned systems:
  - Drones used for surveillance & precision strikes
  - Killer robots
- AI in targeting and decision support: pattern recognition, predictive analytics, and decision support.
- Anti-drone systems: jamming, interception, and destruction

The Economist

Real-World Examples



China to deploy battery-swapping humanoid robots for patrols along Vietnam border

China just signed a US$37m deal to put self-charging humanoid robots on its borders.

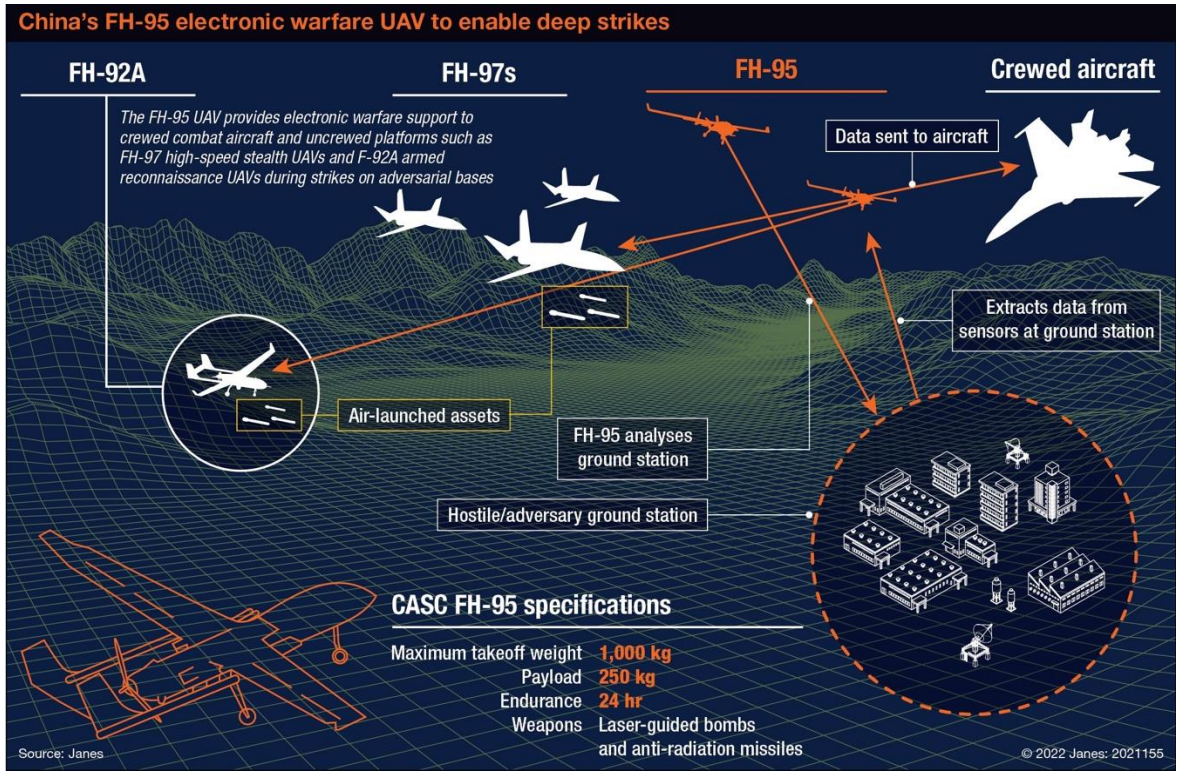
By **Kaif Shaikh** | Nov 26, 2025 08:32 AM EST

GET YOUR NEWS FROM INTERESTING ENGINEERING

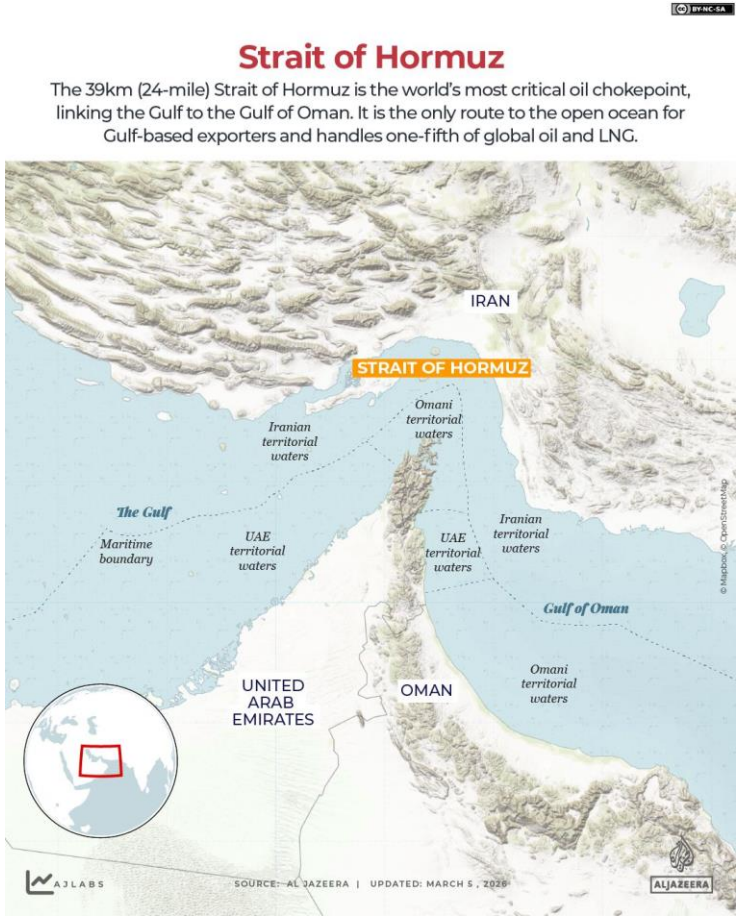
UBTech's advanced Walker S2 brings 24/7 humanoid operation to border. UBTECH/X

Autonomous Weapons Reshape Warfare

- From massed armies to **remote** strike
- **Persistent** surveillance and information dominance
- Blurring the **boundary** between war and peace
- Logistics and supply chains
- **Asymmetric warfare** and decentralized tactics

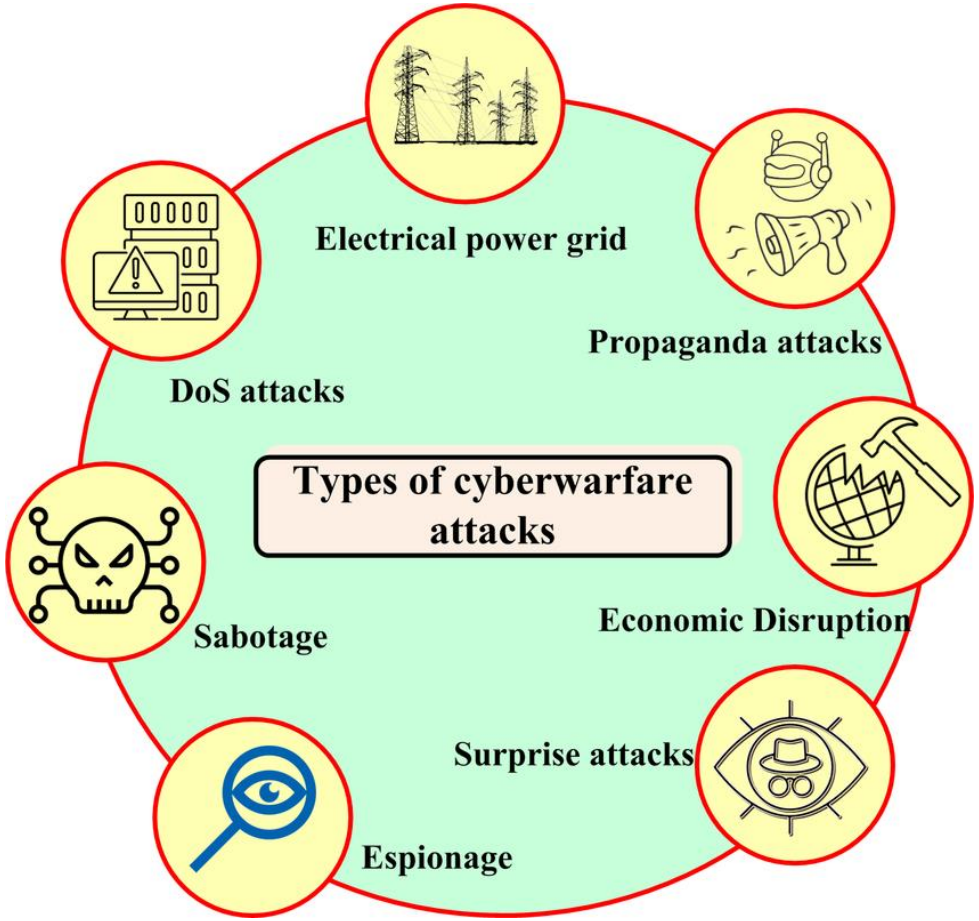


Asymmetric Warfare Between US/Israel and Iran



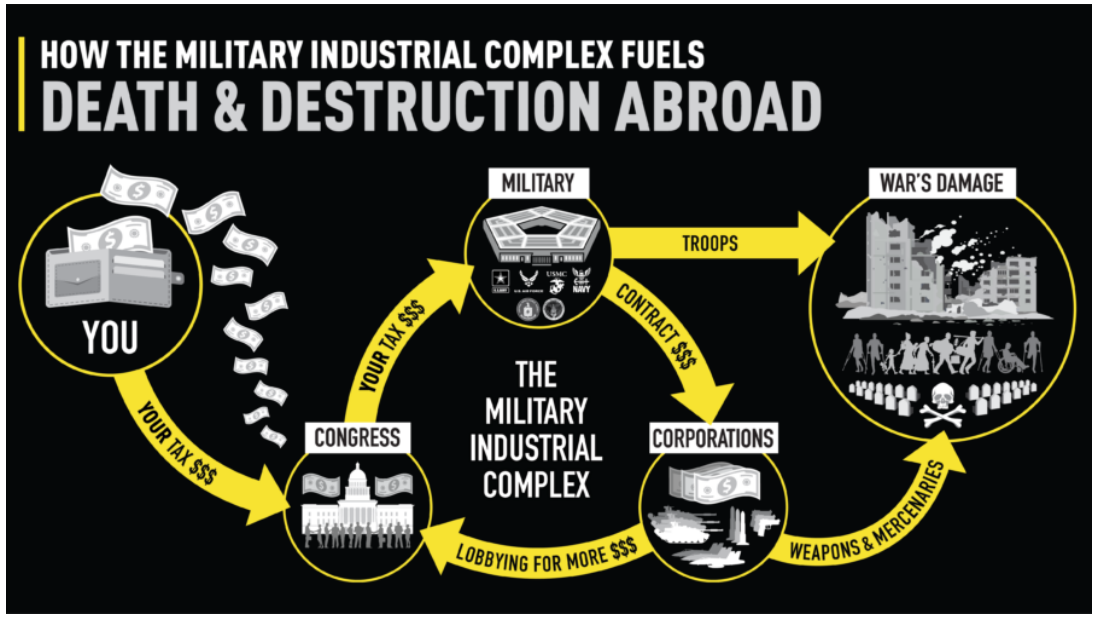
Cyber Warfare

- AI in hacking, malware development, and cyber defense
- Disinformation, deepfakes, and psychological operations
  - Blurred boundaries between propaganda and participation
- **PlayStation** warfare: remote operators and ethical ambiguity



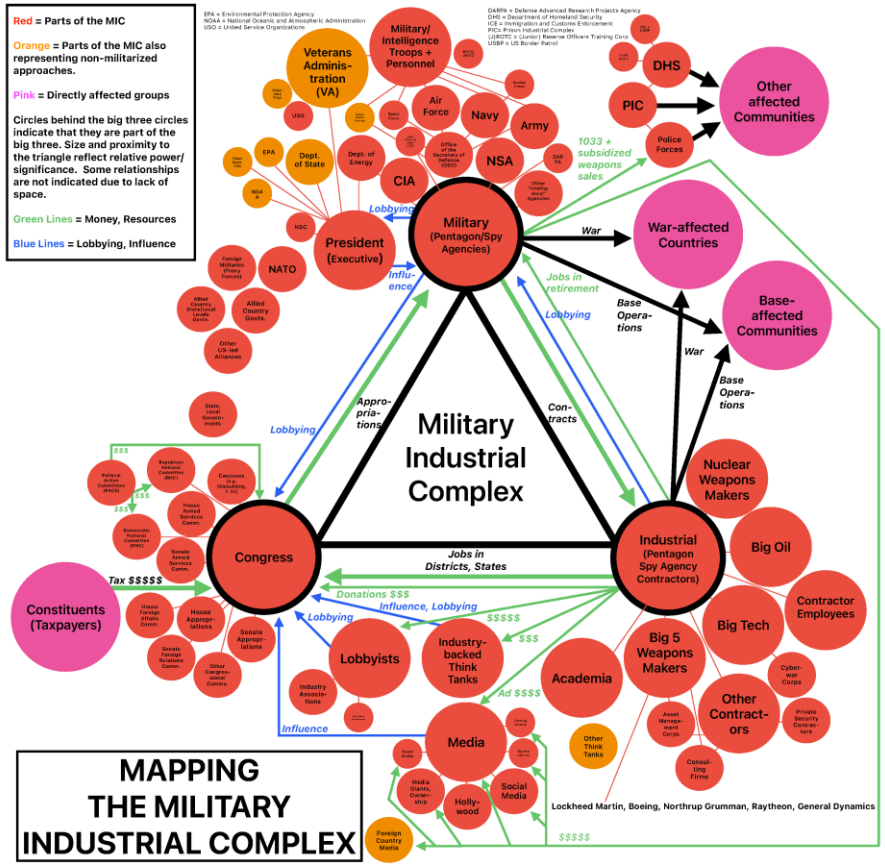
Military-Industrial Complex (MIC)

- Eisenhower's warning (1961)
- Key components: **military** establishment + **private** arms manufacturers and contractors + scientific, technical, and **academic** research institutions
- War **profiteering** is justified in the name of "national security" or even "promoting peace."

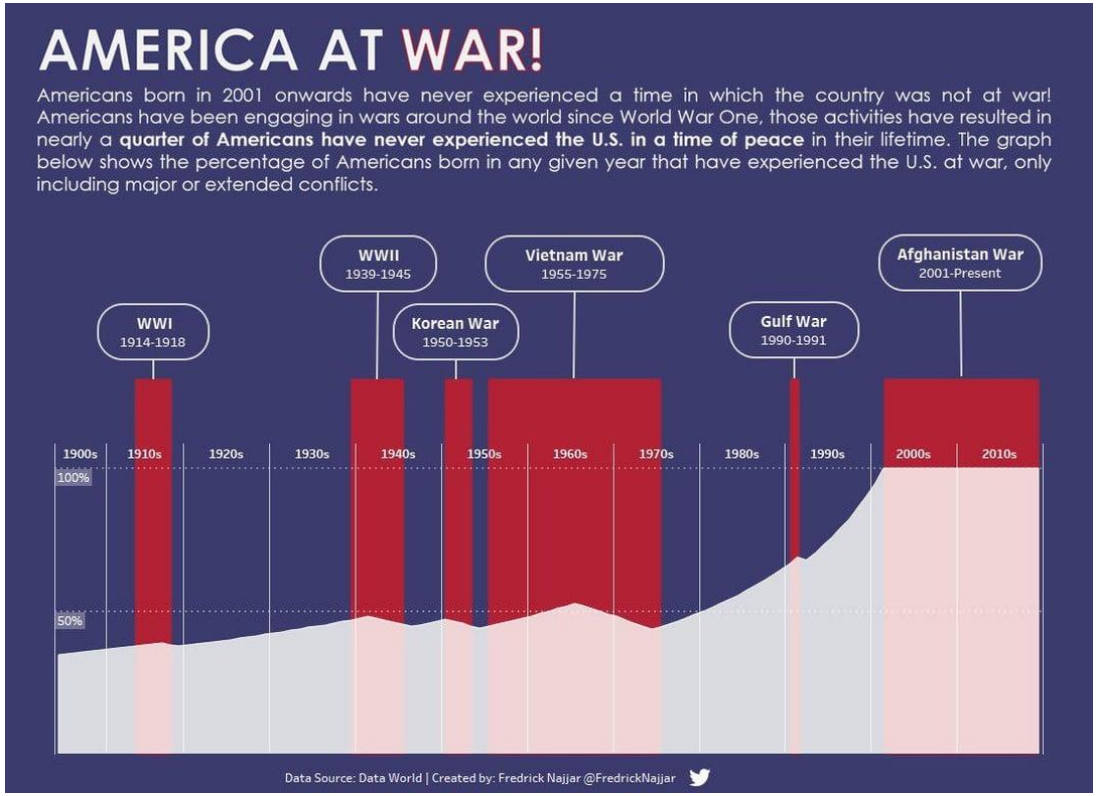


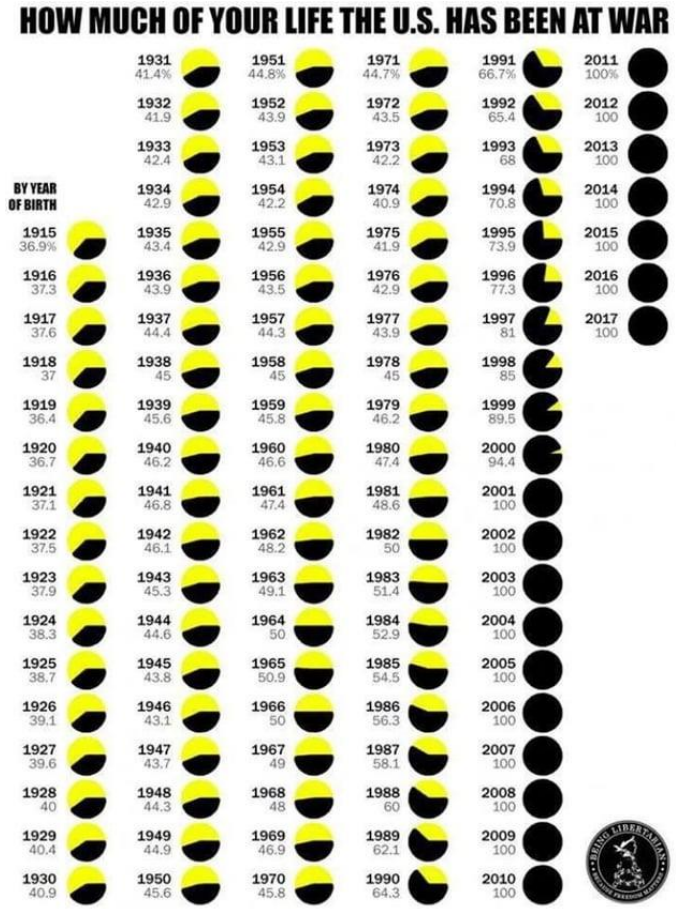
MIC in the US

Purchaser
U.S. Department of Defense
NASA
NATO OTAN
Aerospace & Defense Manufacturing
Cybersecurity & Technology
Naval & Shipbuilding
Engineering, Design & Logistics
Infrastructure & Maintenance
Prime Contractors (Tier 1)
LOCKHEED MARTIN
NORTHROP GRUMMAN
RTX
BAE SYSTEMS
BOEING
L3HARRIS
GENERAL DYNAMICS
Huntington Ingalls Industries
leidos
Major System Subcontractors (Tier 2)
Honeywell
ROLLS ROYCE
TEXTRON
FLUOR
SAIC
PARSONS
BWXT
TRANSDIGM
Microsoft
Jacobs
OSHKOSH
AECOM
Palantir
HEICO
KBR
Specialized Subcontractors (Tier 3)
CROWDSTRIKE
TELEDYNE TECHNOLOGIES
LEONARDO DRS
KRATOS
mercury
AAR
Ducommun
CURTISS-WRIGHT
LOAR GROUP
AV AeroVironment
INTUITIVE MACHINES
TRIUMPH
ADRE HOLDINGS
V2X
HOWMET AEROSPACE
CRANE
MOOG
HEXCEL
PRESTO
SPIRIT AEROSYSTEMS
ROCKETLAB


Red = Parts of the MIC
Orange = Parts of the MIC also representing non-militarized approaches.
Pink = Directly affected groups
Circles behind the big three circles indicate that they are part of the big three. Size and proximity to the triangle reflect relative power/ significance. Some relationships are not indicated due to lack of space.
Green Lines = Money, Resources
Blue Lines = Lobbying, Influence
Military/ Intelligence Troops + Personnel
Veterans Adminis-tration (VA)
Air Force
Navy
Army
NSA
CIA
Dept. of Energy
Dept. of State
President (Executive)
NATO
Military (Pentagon/Spy Agencies)
DHS
PIC
Police Forces
Other affected Communities
War-affected Countries
Base-affected Communities
1033 + subsidized weapons sales
Jobs in retirement
Lobbying
Influ-ence
War
Base Opera-tions
Appro-pria-tions
Con-tracts
Military Industrial Complex
Congress
Jobs in Districts, States
Industrial (Pentagon Spy Agency Contractors)
Nuclear Weapons Makers
Big Oil
Big Tech
Contractor Employees
Big 5 Weapons Makers
Other Contract-ors
Academia
Constituents (Taxpayers)
Tax $$$$$
Donations $$$
Influence, Lobbying
Lobbyists
Industry-backed Think Tanks
Media
Social Media
Hollywood
Influence
Ad $$$$
Lockheed Martin, Boeing, Northrup Grumman, Raytheon, General Dynamics
MAPPING THE MILITARY INDUSTRIAL COMPLEX

The US Has Been in Prolonged Warfare





Silicon Valley and the MIC

- Cold War origins
  - Defense funding (e.g., DARPA) shaped early computing and electronics
  - Close ties between Stanford, startups, and Pentagon research
- Military-**digital**-industrial complex
  - Platforms and algorithms as core **military** infrastructure
  - Blurred line between civilian innovation and militarization

Cases of Commercial AI in Modern Warfare

- Venezuela operation (Jan 2026): Anthropic's Claude, integrated via Palantir's platforms, was used to support military data analysis
- Strike on Iran (Feb 2026-): AI systems were used to support intelligence assessment, target identification, and battlefield simulation

Theory of The State

- *Max Weber*: the state is a polity that claims the **monopoly of the legitimate use of violence** within a given territory.
  - *Charles Tilly* argues states carry out essential activities like war-making and state-making.
  - Legitimacy based on tradition, charisma, or rational-legal authority
- *John Locke*: the goal of the state is the **preservation of property**.
  - *Adam Smith*: provision of **public goods** as a central function of the state
  - Government legitimacy based on the **consent** of the governed (**social contract**); state as the alternative to anarchy (state of nature)

Industrial Policy

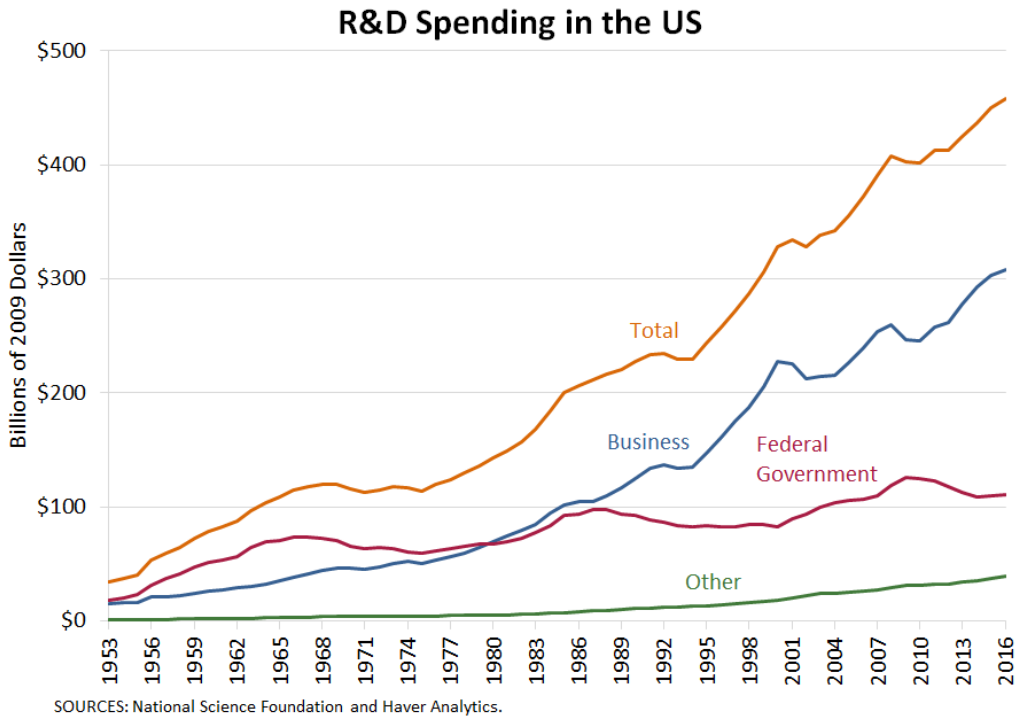
- Liberal-market economies vs. planned economies
- Industrial policy typically builds on comparative advantage.
- Diverse forms of industrial policy

Debate on Industrial Policy

- Alexander Hamilton's infant-industry theory (1790)
- Critiques: industry policy can be undermined by limited knowledge, rent-seeking, and capture by interest groups.
  - "Picking winners?"
- Government failure vs. market failure

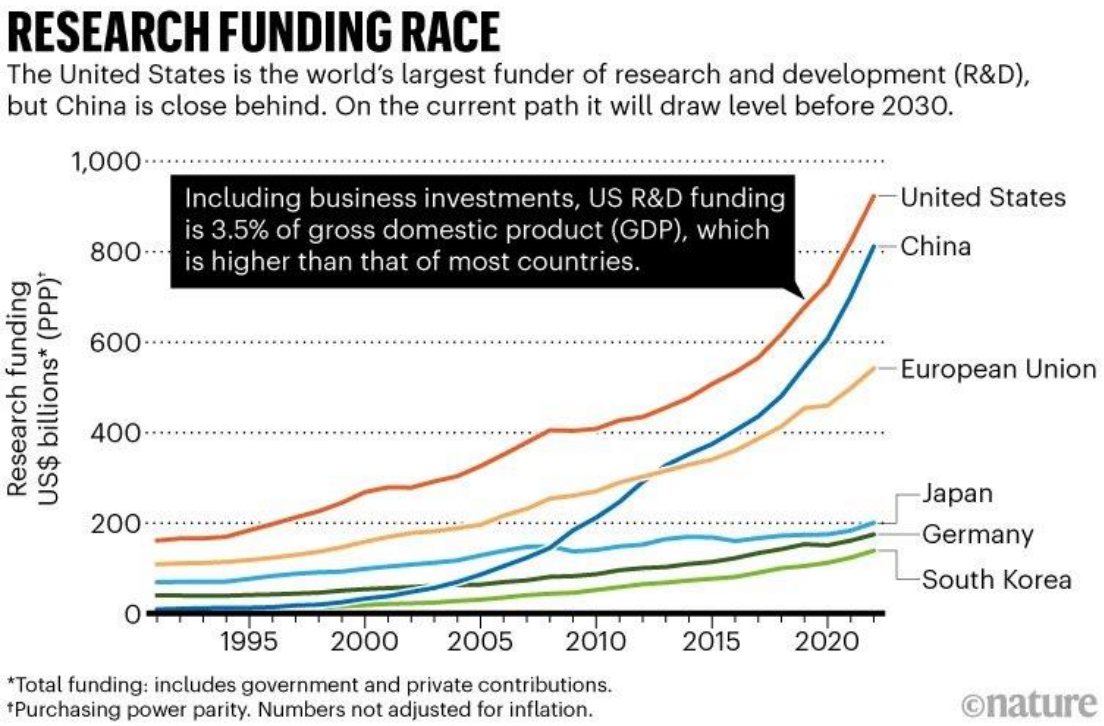
Science & Tech Policy: The US Case

- Postwar Period
  - National Science Foundation (NSF) (1950)
  - National Institutes of Health (NIH) (1946-49)
- Cold War Era
  - DARPA and NASA (1958)
- Post-Cold War
  - Biotechnology, AI, quantum computing, etc.



Science & Tech Policy: The Chinese Case

- Science Policy
  - Institutional reform
  - National Basic Research Program (973 Program) (1997)
  - Talent people policy (2008)
- Technology Policy
  - Market-for-technology strategy
  - National High-tech R&D Program (863 Program) (1986)
  - Made in China 2025 (2015)

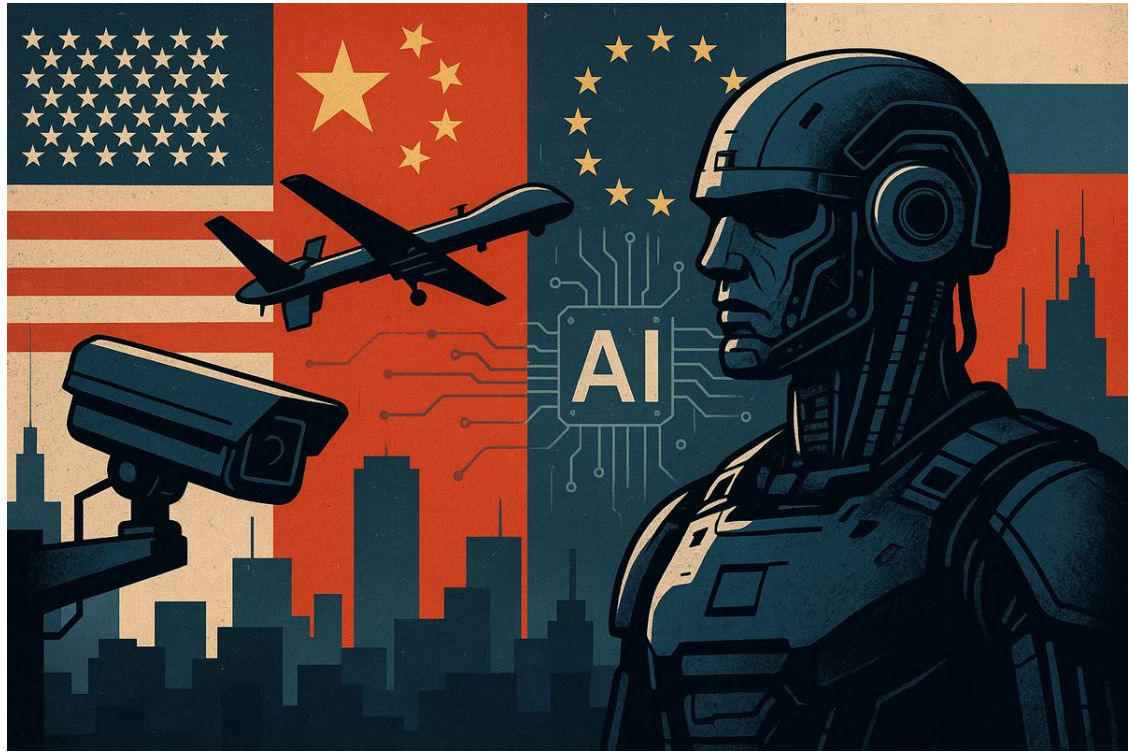


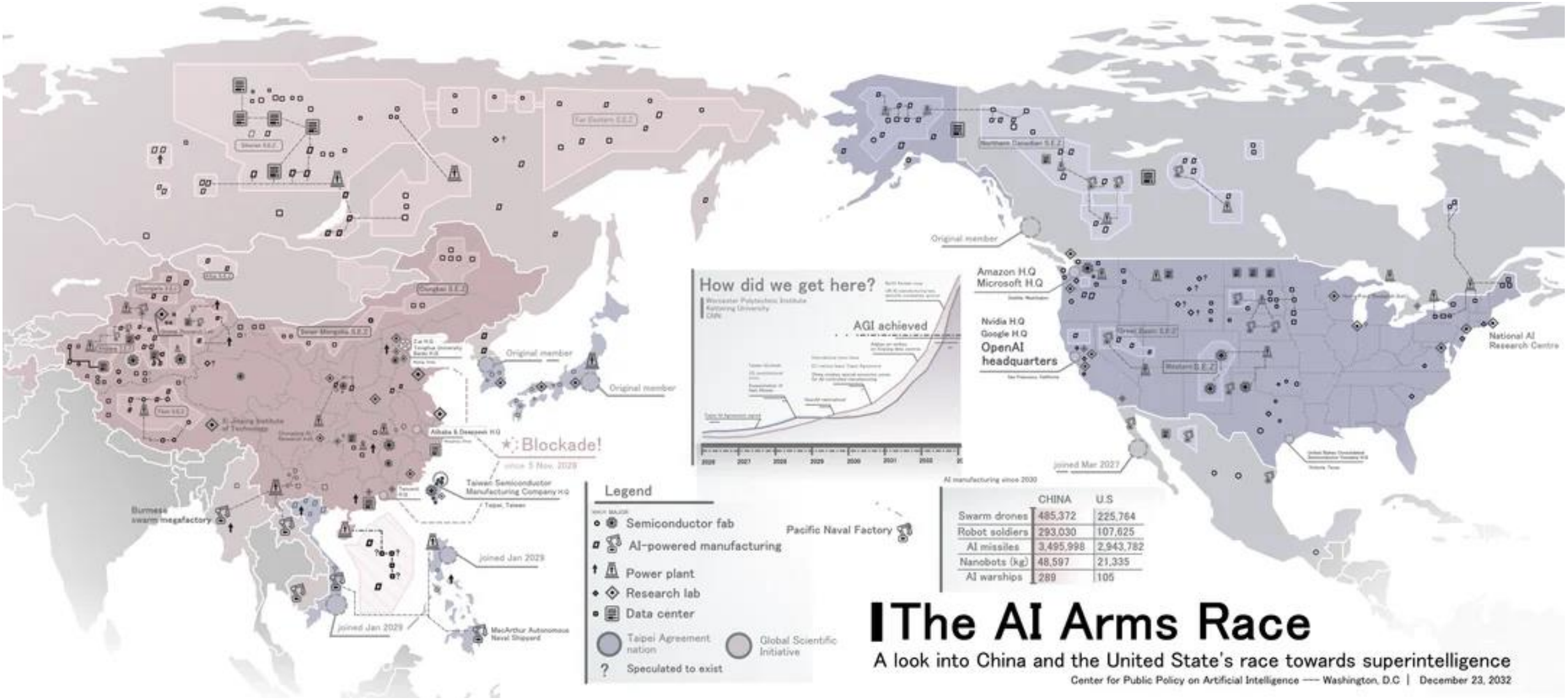
Techno-nationalism

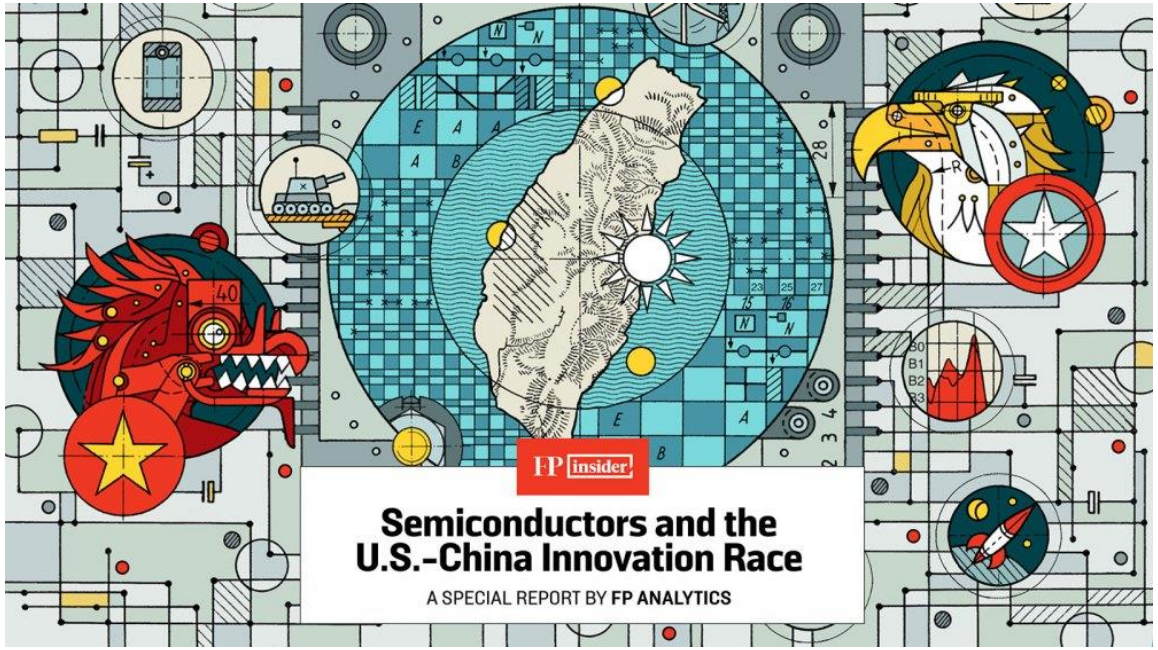
- Technology as a **source** of national power and sovereignty
- Historical roots: mercantilism → Cold War big science → digital era
- Data sovereignty and digital borders: data as a **national resource**
- AI as a site of meaning-making and political imagination

AI Arms Races

- US-China strategic rivalry in AI
- First-mover advantages and escalation dynamics
- Control over AI **compute** capacity: hyperscale data centers as **geopolitical** assets
- Technical standards as geopolitical tools




How did we get here?
AGI achieved
Blockade!
Original member
Original member
Original member
Amazon H.Q
Microsoft H.Q
Nvidia H.Q
Google H.Q
OpenAI headquarters
National AI Research Centre
joined Mar 2027
Taiwan Semiconductor Manufacturing Company H.Q
Burmese swarm megafactory
joined Jan 2029
joined Jan 2029
MacArthur Autonomous Naval Shipyard
Pacific Naval Factory
Legend
Semiconductor fab
AI-powered manufacturing
Power plant
Research lab
Data center
Taipei Agreement nation
Global Scientific Initiative
Speculated to exist
AI manufacturing since 2030
CHINA U.S
Swarm drones 485,372 225,764
Robot soldiers 293,030 107,625
AI missiles 3,495,998 2,943,782
Nanobots (kg) 48,597 21,335
AI warships 289 105
The AI Arms Race
A look into China and the United State's race towards superintelligence
Center for Public Policy on Artificial Intelligence —— Washington, D.C | December 23, 2032

Semiconductor Geopolitics

- Chips as the foundation of AI
- Specialized **chokepoints** in the supply chains
- Technological rivalry between the US and China
  - Export controls, chip embargoes, and technology blockades
  - ASML, Nvidia, Huawei, etc.
  - The Taiwan question (TSMC)